

Translated

From: e-maxx.ru

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Edge connectivity / Vertex connectivity

Definition

Given an undirected graph G with n vertices and m edges. Both the edge connectivity and the vertex connectivity are characteristics describing the graph.

Edge connectivity

The **edge connectivity** λ of the graph G is the minimum number of edges that need to be deleted, such that the graph G gets disconnected.

For example an already disconnected graph has an edge connectivity of 0 , a connected graph with at least one bridge has an edge connectivity of 1 , and a connected graph with no bridges has an edge connectivity of at least 2 .

We say that a set S of edges **separates** the vertices s and t , if, after removing all edges in S from the graph G , the vertices s and t end up in different connected components.

It is clear, that the edge connectivity of a graph is equal to the minimum size of such a set separating two vertices s and t , taken among all possible pairs (s, t) .

Vertex connectivity

The **vertex connectivity** κ of the graph G is the minimum number of vertices that need to be deleted, such that the graph G gets disconnected.

For example an already disconnected graph has the vertex connectivity 0 , and a connected graph with an articulation point has the vertex connectivity 1 . We define that a complete graph has the vertex connectivity $n-1$. For all other graphs the vertex connectivity doesn't exceed $n-2$, because you can find a pair of vertices which are not connected by an edge, and remove all other $n-2$ vertices.

We say that a set T of vertices **separates** the vertices s and t , if, after removing all vertices in T from the graph G , the vertices end up in different connected components.